

Thrall's problem for two rows

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How this project started

- ▶ **KRIMS 2026 Winter School**

Symmetric Functions and Group Representations
Kangwon National University, February 23–27, 2026.

- ▶ **Thrall's problem for two rows**

JiSun Huh, Woo-Seok Jung, Jang Soo Kim, Meesue Yoo
arXiv:2605.17880, May 2026.

Further study on background

- ▶ **Representation Theory**

Lecture videos from my course in 2016.

- ▶ Richard P. Stanley, **Enumerative Combinatorics, Vol. 2**

Chapter 7: symmetric functions, Schur functions, and tableaux.

- ▶ William Fulton and Joe Harris,
Representation Theory: A First Course

Representations, characters, and symmetric groups.

Outline

Group representations

Symmetric functions

Higher Lie modules

Thrall's problem

Block-major indices and refined Thrall subsets

Jeu de taquin slides and rectification

Main result: the two-row case $\lambda = (n, n)$

The bijection ξ

Section 1

Group representations

Basic definitions

- ▶ $[a, b] = \{a, a + 1, \dots, b\}$, $[n] = [1, n] = \{1, 2, \dots, n\}$.
- ▶ **Group**: a set G with a product $(a, b) \mapsto ab$:
 - ▶ **Associativity**: $(ab)c = a(bc)$ for all $a, b, c \in G$.
 - ▶ **Identity**: $e \in G$, $ea = ae = a$ for all $a \in G$.
 - ▶ **Inverses**: Each $a \in G$ has $a^{-1} \in G$ with $aa^{-1} = a^{-1}a = e$.
- ▶ **Permutation**: a bijection $\sigma : [n] \rightarrow [n]$.
- ▶ **Symmetric group** $\mathfrak{S}_n$: all permutations, under composition.
- ▶ **General linear group** $\text{GL}(V)$: invertible linear maps $V \rightarrow V$, under composition; V finite-dimensional over $\mathbb{C}$.
- ▶ **Action** on X : a map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x$, with

$$e \cdot x = x, \quad (gh) \cdot x = g \cdot (h \cdot x) \quad \text{for all } g, h \in G, x \in X$$

Group representations and modules

- **Representation**: an action of G on a complex vector space W by linear maps:

$$g \cdot (au + bv) = a(g \cdot u) + b(g \cdot v)$$

for all $g \in G$, $u, v \in W$, and $a, b \in \mathbb{C}$.

- Equivalently, $\rho : G \rightarrow \mathrm{GL}(W)$ with

$$\rho(e) = \mathrm{id}_W, \quad \rho(gh) = \rho(g)\rho(h) \quad \text{for all } g, h \in G,$$

$$\rho(g)(w) = g \cdot w.$$

- **G -module**: W together with this action.

Irreducible modules and decomposition

- ▶ **Submodule**: a subspace $U \subseteq W$ preserved by G
 - ▶ $g \cdot u \in U$ for all $g \in G, u \in U$.
- ▶ **Irreducible**: $W \neq 0$, with only submodules 0 and W .

Theorem (Maschke)

- ▶ G finite, W a finite-dimensional complex G -module:

$$W \cong \bigoplus_{i=1}^r U_i^{\oplus m_i},$$

- ▶ U_i : pairwise nonisomorphic irreducible G -modules.
- ▶ **Multiplicity** m_i : number of copies of U_i ; uniquely determined.
- ▶ Also holds for finite-dimensional **polynomial $\mathrm{GL}(V)$ -modules** over $\mathbb{C}$ (entries of $\rho(g)$ are polynomials in entries of g)

Characters

- ▶ W : finite-dimensional complex G -module, $\rho : G \rightarrow \mathrm{GL}(W)$.
- ▶ The **character** of W :

$$\chi_W : G \rightarrow \mathbb{C}, \quad \chi_W(g) = \mathrm{tr}(\rho(g)).$$

- ▶ Direct sums give sums of characters:

$$\chi_{U \oplus W}(g) = \chi_U(g) + \chi_W(g).$$

Decomposing a module decomposes its character

$$W \cong \bigoplus_{i=1}^r U_i^{\oplus m_i} \implies \chi_W = \sum_{i=1}^r m_i \chi_{U_i}.$$

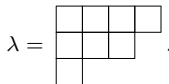
- ▶ **Converse**: holds for completely reducible modules (finite groups; polynomial $\mathrm{GL}(V)$ -modules over $\mathbb{C}$).

Section 2

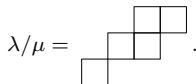
Symmetric functions

Partitions and Young diagrams

- ▶ **Partition**: a weakly decreasing sequence $\lambda = (\lambda_1, \dots, \lambda_\ell)$ of positive integers.
- ▶ λ_i : a **part**.
- ▶ $|\lambda| = \lambda_1 + \dots + \lambda_\ell = n$: partition of n , written $\lambda \vdash n$.
- ▶ **Young diagram** of $\lambda = (4, 3, 1)$:



- ▶ **Skew shape** λ/μ : remove μ 's cells from λ ($\mu \subseteq \lambda$).
- ▶ Example: $\lambda = (4, 3, 1)$, $\mu = (2, 1)$.



Semistandard and standard Young tableaux

- **Semistandard Young tableau** (SSYT): fill λ with positive integers such that rows weakly increasing, columns strictly increasing.

1	1	2	3
2	2	3	
4			

- **Standard Young tableau** (SYT): an SSYT using $1, 2, \dots, n$ once each.

1	2	3	8
4	6	7	
5			

- $\text{SYT}(\lambda)$: SYT of shape λ
- $\text{SYT}(n)$: SYT of size n .
- Irreducible complex $\mathfrak{S}_n$ -modules S^λ , $\lambda \vdash n$
- $\dim S^\lambda = |\text{SYT}(\lambda)|$.
- Irreducible polynomial $\text{GL}(V)$ -modules V_λ : λ has at most $N = \dim V$ parts.
- $\dim V_\lambda = |\text{SSYT}_N(\lambda)|$: SSYT with entries in $[N]$.

Symmetric polynomials and symmetric functions

- **Symmetric polynomial**: unchanged by permuting variables:
for every $\sigma \in \mathfrak{S}_m$,

$$f(x_1, \dots, x_m) = f(x_{\sigma(1)}, \dots, x_{\sigma(m)}).$$

- Examples in two variables:

symmetric	not symmetric
$x_1 + x_2$	$x_1 + 2x_2$
x_1x_2	$x_1^2x_2$

- **Symmetric function**: symmetric polynomial with infinite variables $x = (x_1, x_2, \dots)$
- Λ_n : symmetric functions homogeneous of degree n .

Bases of symmetric functions

- ▶ Let $\lambda = (\lambda_1, \dots, \lambda_\ell)$.
- ▶ **Elementary symmetric functions:**

$$e_\lambda := e_{\lambda_1} e_{\lambda_2} \cdots ,$$

where

$$e_d := \sum_{i_1 < \cdots < i_d} x_{i_1} \cdots x_{i_d}.$$

- ▶ **Homogeneous symmetric functions:**

$$h_\lambda := h_{\lambda_1} h_{\lambda_2} \cdots ,$$

where

$$h_d := \sum_{i_1 \leq \cdots \leq i_d} x_{i_1} \cdots x_{i_d}.$$

Example

- ▶ Two variables x_1, x_2 , $\lambda = (2, 1)$:

$$e_{(2,1)} = e_2 e_1 = x_1 x_2 (x_1 + x_2) = x_1^2 x_2 + x_1 x_2^2,$$

$$\begin{aligned} h_{(2,1)} &= h_2 h_1 = (x_1^2 + x_1 x_2 + x_2^2)(x_1 + x_2) \\ &= x_1^3 + 2x_1^2 x_2 + 2x_1 x_2^2 + x_2^3. \end{aligned}$$

Schur functions

► **Schur functions:**

$$s_{\lambda}(x) = \sum_{T \in \text{SSYT}(\lambda)} x_T,$$

$$T = \begin{array}{|c|c|c|c|} \hline 1 & 1 & 2 & 3 \\ \hline 2 & 2 & 3 & \\ \hline 4 & & & \\ \hline \end{array} \quad x_T = x_1^2 x_2^3 x_3^2 x_4.$$

► Shape $(2, 1)$, entries in $\{1, 2\}$: exactly two SSYT.

$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline 2 & \\ \hline \end{array} \quad \text{and} \quad \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 2 & \\ \hline \end{array} \quad \Longrightarrow \quad s_{(2,1)}(x_1, x_2) = x_1^2 x_2 + x_1 x_2^2.$$

► Three bases of Λ_n :

$$\{h_{\lambda} : \lambda \vdash n\}, \quad \{e_{\lambda} : \lambda \vdash n\}, \quad \{s_{\lambda} : \lambda \vdash n\}$$

Section 3

Higher Lie modules

Tensor products of vector spaces

Definition

- ▶ U, W : vector spaces over $\mathbb{C}$.
- ▶ **Tensor product** $U \otimes W$: finite linear combinations of symbols $u \otimes w$ subject only to **bilinearity**:

$$(u + u') \otimes w = u \otimes w + u' \otimes w,$$

$$u \otimes (w + w') = u \otimes w + u \otimes w',$$

$$a(u \otimes w) = (au) \otimes w = u \otimes (aw),$$

where $u, u' \in U$, $w, w' \in W$, and $a \in \mathbb{C}$.

- ▶ Bases $u_1, \dots, u_r$ of U , $w_1, \dots, w_s$ of W :

$$\{u_i \otimes w_j : 1 \leq i \leq r, 1 \leq j \leq s\}$$

is a basis of $U \otimes W$

- ▶ **Tensor power**: $V^{\otimes m} = \underbrace{V \otimes \dots \otimes V}_{m \text{ factors}}$.

Example

- ▶ V : basis a, b . Then $V \otimes V$: basis $a \otimes a$, $a \otimes b$, $b \otimes a$, $b \otimes b$.
- ▶ Expand by bilinearity:

$$(a + b) \otimes (a - b) = a \otimes a - a \otimes b + b \otimes a - b \otimes b.$$

Tensor algebra: a vector space of words

- ▶ $V = \mathbb{C}^N$; basis $e_1, \dots, e_N = \text{letters}$.
- ▶ $V^{\otimes m}$: basis of all words of length m :

$$e_{i_1} \otimes \dots \otimes e_{i_m} \quad \text{is written as} \quad e_{i_1} \dots e_{i_m}.$$

- ▶ Vectors = linear combinations of words; $\dim V^{\otimes m} = N^m$.
- ▶ Example: V with basis a, b .

$$V^{\otimes 2} = \text{span}\{aa, ab, ba, bb\}.$$

Order matters: $ab \neq ba$.

- ▶ **Tensor algebra**: all word lengths.

$$T(V) = \mathbb{C} \oplus V \oplus V^{\otimes 2} \oplus \dots.$$

- ▶ $\oplus$: separate components by length; $\mathbb{C}$: empty word 1.

Multiplication = concatenation, extended linearly

$$(ab)a = aba, \quad (a + b)a = aa + ba.$$

Free Lie algebra: build expressions using brackets

- **Commutator bracket** in $T(V)$:

$$[u, v] = uv - vu.$$

- Examples:

$$[a, b] = ab - ba, \quad [a, [a, b]] = aab - 2aba + baa.$$

- $\mathcal{L}_m(V)$: span of brackets using exactly m letters

$$\mathcal{L}_1(V) = V,$$

$$\mathcal{L}_2(V) = \text{span}\{[e_i, e_j] : i, j \in [N]\},$$

$$\mathcal{L}_3(V) = \text{span}\{[e_i, [e_j, e_k]] : i, j, k \in [N]\},$$

$$\mathcal{L}_m(V) \subseteq V^{\otimes m}.$$

- **Free Lie algebra**

$$\text{Lie}(V) = \bigoplus_{m \geq 1} \mathcal{L}_m(V)$$

Symmetric powers: unordered products

- **Symmetric power** $\text{Sym}^r(W)$: $W^{\otimes r}$ with order ignored.
- $u \odot v = v \odot u$.
- Example: W with basis u, v .

space	basis	dimension
$W \otimes W$	$u \otimes u, u \otimes v, v \otimes u, v \otimes v$	4
$\text{Sym}^2(W)$	$u \odot u, u \odot v, v \odot v$	3

- Basis $w_1, \dots, w_d$ of W gives a basis of $\text{Sym}^r(W)$:

$$\{w_{j_1} \odot \dots \odot w_{j_r} : 1 \leq j_1 \leq \dots \leq j_r \leq d\}.$$

- $\dim \text{Sym}^r(W) = \binom{d+r-1}{r}$: multisets of size r from d basis vectors.
- $\text{Sym}^0(W) = \mathbb{C}$, $\text{Sym}^1(W) = W$.
- **Symmetric algebra**:

$$\text{Sym}(W) = \bigoplus_{r \geq 0} \text{Sym}^r(W).$$

Higher Lie modules

► $\lambda = (\lambda_1, \dots, \lambda_\ell); m_i(\lambda)$: number of parts equal to i .

► **Higher Lie module:**

$$L_\lambda := \mathcal{L}_\lambda(V) := \text{span}\{u_1 \odot \dots \odot u_\ell : u_j \in \mathcal{L}_{\lambda_j}(V)\} \subseteq \text{Sym}^\ell(\text{Lie}(V)).$$

► Equivalently, as $\text{GL}(V)$ -modules:

$$L_\lambda \cong \bigotimes_{i \geq 1} \text{Sym}^{m_i(\lambda)}(\mathcal{L}_i(V)).$$

Example

► $\lambda = (3, 2, 2, 1) \vdash 8$: $m_1 = 1, m_2 = 2, m_3 = 1$. Reordering tensor factors:

$$L_{(3,2,2,1)} \cong \underbrace{\mathcal{L}_3(V)}_{\text{one length-3 bracket}} \otimes \underbrace{\text{Sym}^2(\mathcal{L}_2(V))}_{\text{two length-2 brackets}} \otimes \underbrace{V}_{\text{one letter}}.$$

► For $a, b, c, d \in V$:

$$[a, [a, b]] \odot [a, b] \odot [c, d] \odot a \quad \longleftrightarrow \quad [a, [a, b]] \otimes ([a, b] \odot [c, d]) \otimes a.$$

$$L_{(n)} = \mathcal{L}_n(V), \quad L_{(1^n)} = \text{Sym}^n(V), \quad L_{(2,1)} \cong \mathcal{L}_2(V) \otimes V.$$

► Abbreviation: $L_n = L_{(n)}$.

Why these spaces?

Poincaré–Birkhoff–Witt (PBW) theorem

- ▶ Vector-space isomorphism preserving total word length:

$$T(V) \cong \text{Sym}(\text{Lie}(V)).$$

- ▶ Group bracket products by their lengths:

$$V^{\otimes n} \cong \bigoplus_{\lambda \vdash n} L_{\lambda}.$$

- ▶ $n = 2$: partitions $(1, 1)$, (2) .

$$V^{\otimes 2} \cong \underbrace{\text{Sym}^2(V)}_{L_{(1,1)}} \oplus \underbrace{\mathcal{L}_2(V)}_{L_{(2)}}.$$

- ▶ Example:

$$ab = \underbrace{\frac{ab + ba}{2}}_{\text{symmetric part}} + \underbrace{\frac{ab - ba}{2}}_{\text{bracket part}}.$$

What does “module” mean here?

- ▶ $\mathrm{GL}(V)$: invertible linear maps $V \rightarrow V$.
- ▶ **Representation** / **$\mathrm{GL}(V)$ -module**: a vector space with a linear action.
- ▶ Action: apply to each letter.

$$g(v_1 \cdots v_m) = (gv_1) \cdots (gv_m), \quad g([u, v]) = [g(u), g(v)].$$

- ▶ Preserves bracket lengths; extends to symmetric and tensor products.
- ▶ Hence each L_λ is a $\mathrm{GL}(V)$ -module.

Example

- ▶ $g(a) = a + b, g(b) = b$:

$$g(aa) = aa + ab + ba + bb, \quad g([a, b]) = [a + b, b] = [a, b].$$

- ▶ **Polynomial module**: action matrix entries are polynomials in entries of g .
- ▶ PBW decomposition respects the action.

A polynomial module: an explicit matrix example

- V : basis a, b . Invertible matrix:

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{acts by} \quad g(a) = \alpha a + \gamma b, \quad g(b) = \beta a + \delta b.$$

- $\mathcal{L}_2(V)$: one-dimensional, basis $[a, b]$.

$$g([a, b]) = [\alpha a + \gamma b, \beta a + \delta b] = (\alpha\delta - \beta\gamma)[a, b].$$

Action matrix: $(\alpha\delta - \beta\gamma)$; polynomial in entries of g .

- $\text{Sym}^2(V)$: ordered basis $a \odot a, a \odot b, b \odot b$.

$$g(a \odot a) = (\alpha a + \gamma b) \odot (\alpha a + \gamma b) = \alpha^2(a \odot a) + 2\alpha\gamma(a \odot b) + \gamma^2(b \odot b).$$

- Coefficients give the first column:

$$\begin{pmatrix} \alpha^2 & \alpha\beta & \beta^2 \\ 2\alpha\gamma & \alpha\delta + \beta\gamma & 2\beta\delta \\ \gamma^2 & \gamma\delta & \delta^2 \end{pmatrix}.$$

Other columns: $g(a \odot b), g(b \odot b)$.

Why “polynomial”?

- Every action matrix entry: degree-2 polynomial in g 's entries $\alpha, \beta, \gamma, \delta$.

Characters: record how diagonal matrices act

- ▶ $g = \text{diag}(x_1, \dots, x_N)$
- ▶ $g(e_i) = x_i e_i$
- ▶ The **character** of a polynomial module W for diagonal matrices:

$$\text{ch}(W)(x_1, \dots, x_N) := \text{tr}(\text{matrix describing the action of } g \text{ on } W).$$

- ▶ Each letter contributes its variable:

$$g(e_1 e_2) = x_1 x_2 e_1 e_2, \quad g([e_1, e_2]) = x_1 x_2 [e_1, e_2].$$

W	basis vectors	$\text{ch}(W)$
V	e_i	$\sum_i x_i = h_1$
$\text{Sym}^2(V)$	$e_i \odot e_j \ (i \leq j)$	$\sum_{i \leq j} x_i x_j = h_2$
$\mathcal{L}_2(V)$	$[e_i, e_j] \ (i < j)$	$\sum_{i < j} x_i x_j = e_2$

- ▶ Permute the x_i 's = relabel the basis of V : conjugate action matrices, same trace.
- ▶ $\text{ch}(W)(x_1, \dots, x_N)$: a **symmetric polynomial**.
- ▶ If $x_1 = \dots = x_N = 1$, then $\text{ch}(W)(1, \dots, 1) = \dim W$.

Decomposition and Schur expansion

Theorem

- ▶ $\dim_{\mathbb{C}} V = N \geq n$.
- ▶ *Irreducible polynomial $\mathrm{GL}(V)$ -modules of degree n : $V_{\mu}, \mu \vdash n$.*
- ▶ $\mathrm{ch}(V_{\mu}) = s_{\mu}(x_1, \dots, x_N)$.

Corollary

- ▶ $\dim_{\mathbb{C}} V = N \geq n, \lambda \vdash n$:

$$L_{\lambda} \cong \bigoplus_{\mu \vdash n} V_{\mu}^{\oplus c_{\lambda, \mu}} \iff \mathrm{ch}(L_{\lambda}) = \sum_{\mu \vdash n} c_{\lambda, \mu} s_{\mu}.$$

Section 4

Thrall's problem

Thrall's problem (1942)

- ▶ Given $\lambda \vdash n$, find a combinatorial interpretation of $c_{\lambda, \mu}$.

$$\text{ch}(L_\lambda) = \sum_{\mu \vdash n} c_{\lambda, \mu} s_\mu.$$

- ▶ Known cases:

- ▶ **One column** ($\lambda = (1^n)$):

$$\text{ch}(L_{(1^n)}) = s_n.$$

- ▶ **Two columns** ($\lambda = (2^n)$):

$$\text{ch}(L_{(2^n)}) = \sum_{\substack{\mu \vdash 2n \\ \text{each column has even length}}} s_\mu.$$

- ▶ **One row** ($\lambda = (n)$, Klyachko 1974 / Kraśkiewicz–Weyman 1987):

$$\text{ch}(L_n) = \sum_{\mu \vdash n} |\{T \in \text{SYT}(\mu) : \text{maj}(T) \equiv 1 \pmod{n}\}| s_\mu.$$

- ▶ **Our results:**

- ▶ distinct parts
- ▶ two equal rows (n, n)
- ▶ each part > 2 appearing at most twice.

Descents and major index

- $T \in \text{SYT}(n)$; **descent set**:

$$\text{Des}(T) = \{ i \in [n-1] : i+1 \text{ lies in a strictly lower row than } i \text{ in } T \}.$$

- **Major index**:

$$\text{maj}(T) = \sum_{i \in \text{Des}(T)} i.$$

- **Example**:

$$T = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 6 \\ \hline 3 & 5 & 8 & \\ \hline 7 & & & \\ \hline \end{array},$$

$$\text{Des}(T) = \{2, 4, 6\} \text{ and } \text{maj}(T) = 2 + 4 + 6 = 12.$$

(λ, μ) -Thrall subsets

- ▶ PBW theorem and Schur–Weyl duality:

$$\bigoplus_{\lambda \vdash n} L_\lambda \cong V^{\otimes n} \cong \bigoplus_{\mu \vdash n} V_\mu^{\oplus |\text{SYT}(\mu)|},$$

V_μ : irreducible $\text{GL}(V)$ -module indexed by μ .

- ▶ Characters:

$$\sum_{\lambda \vdash n} \text{ch}(L_\lambda) = \sum_{\lambda \vdash n} \sum_{\mu \vdash n} c_{\lambda, \mu} s_\mu = \sum_{\mu \vdash n} |\text{SYT}(\mu)| s_\mu.$$

- ▶ Fix $\mu \vdash n$:

$$\sum_{\lambda \vdash n} c_{\lambda, \mu} = |\text{SYT}(\mu)|.$$

- ▶ Hence, for $\lambda, \mu \vdash n$:

$$c_{\lambda, \mu} \leq |\text{SYT}(\mu)|.$$

Definition

- ▶ (λ, μ) -**Thrall subset**: subset of $\text{SYT}(\mu)$ of size $c_{\lambda, \mu}$.
- ▶ **Goal**: Find (λ, μ) -Thrall subsets.

Section 5

Block-major indices and refined Thrall subsets

Block-major indices

- ▶ $\lambda = (\lambda_1, \dots, \lambda_r) \vdash n$.
- ▶ Block i :

$$B_{\lambda,i} = [\lambda_1 + \dots + \lambda_{i-1} + 1, \lambda_1 + \dots + \lambda_i].$$

Definition

- ▶ $T \in \text{SYT}(n)$: descents within block i , shifted to start at 1.

$$\text{Des}_{\lambda,i}(T) = \{d - (\lambda_1 + \dots + \lambda_{i-1}) : d \in \text{Des}(T), d, d+1 \in B_{\lambda,i}\},$$

$$\text{maj}_{\lambda,i}(T) = \sum_{d \in \text{Des}_{\lambda,i}(T)} d.$$

- ▶ $\text{maj}_{\lambda,i}(T)$: **i -th block-major index**.

Refined Thrall subsets

Definition

- $\lambda, \mu \vdash n$:

$$\text{SYT}_\lambda(\mu) := \{ T \in \text{SYT}(\mu) : \text{maj}_{\lambda,i}(T) \equiv 1 \pmod{\lambda_i} \text{ for all } i \}.$$

- **Refined (λ, μ) -Thrall subset**: subset of $\text{SYT}_\lambda(\mu)$ of size $c_{\lambda,\mu}$.

Example

- $\lambda = (5, 3), \mu = (4, 3, 1)$

- $B_{\lambda,1} = \{1, 2, 3, 4, 5\}, B_{\lambda,2} = \{6, 7, 8\}.$

- $T = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 6 \\ \hline 3 & 5 & 8 & \\ \hline 7 & & & \\ \hline \end{array} \in \text{SYT}((4, 3, 1)); \text{Des}(T) = \{2, 4, 6\}.$

- Block 2 shifted by $\lambda_1 = 5$:

$$\text{maj}_{\lambda,1}(T) = 2 + 4 = 6 \equiv 1 \pmod{5}$$

$$\text{maj}_{\lambda,2}(T) = 6 - 5 = 1 \equiv 1 \pmod{3}$$

- $T \in \text{SYT}_{(5,3)}((4, 3, 1)).$

Distinct partitions

Theorem

- ▶ $\lambda \vdash n$ with *distinct parts*:

$$\text{ch}(L_\lambda) = \sum_{\mu \vdash n} |\text{SYT}_\lambda(\mu)| s_\mu.$$

- ▶ $\text{SYT}_\lambda(\mu)$: a (λ, μ) -*Thrall subset*.

Theorem

- ▶ $\lambda, \mu \vdash n$:

$$c_{\lambda, \mu} \leq |\text{SYT}_\lambda(\mu)|.$$

- ▶ *Equality for all μ if and only if λ has distinct parts.*

Corollary

- ▶ *Refined (λ, μ) -Thrall subsets always exist.*

Example

- ▶ $\lambda = (2, 1)$.
- ▶ Conditions: $\text{maj}_{\lambda,1} \equiv 1 \pmod{2}$, $\text{maj}_{\lambda,2} \equiv 1 \pmod{1}$.

μ	T	$\text{Des}(T)$	$\text{maj}_{\lambda,1}$	in $\text{SYT}_{(2,1)}(\mu)$?				
(3)	<table><tr><td>1</td><td>2</td><td>3</td></tr></table>	1	2	3	$\emptyset$	0	no	
1	2	3						
(2, 1)	<table><tr><td>1</td><td>2</td></tr><tr><td>3</td><td></td></tr></table>	1	2	3		$\{2\}$	0	no
1	2							
3								
(2, 1)	<table><tr><td>1</td><td>3</td></tr><tr><td>2</td><td></td></tr></table>	1	3	2		$\{1\}$	1	yes
1	3							
2								
(1, 1, 1)	<table><tr><td>1</td></tr><tr><td>2</td></tr><tr><td>3</td></tr></table>	1	2	3	$\{1, 2\}$	1	yes	
1								
2								
3								

- ▶ Result:

$$\text{ch}(L_{(2,1)}) = s_{(2,1)} + s_{(1,1,1)}.$$

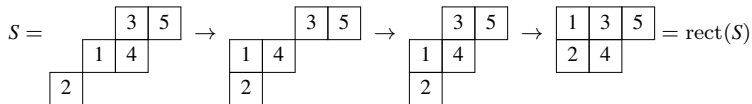
Section 6

Jeu de taquin slides and rectification

Jeu de taquin and rectification

- ▶ S : a skew SYT.
- ▶ **Jeu de taquin (jdt) slide**:
 - ▶ Start with a hole at an inner corner.
 - ▶ Move the smaller right/lower neighbor into the hole; repeat.
 - ▶ Delete the hole at an outer corner.
- ▶ **Rectification** $\text{rect}(S)$: repeat jdt slides to obtain a straight-shape SYT.
- ▶ Schützenberger: $\text{rect}(S)$ is independent of slide order.

Example



Restriction T_I and rectification T^I

- ▶ $T \in \text{SYT}(\mu)$, $\mu \vdash n$; interval $I \subseteq [n]$.
- ▶ T_I : keep entries in I .
- ▶ $T^I := \text{rect}(\text{st}(T_I))$: relabel increasingly by $1, \dots, |I|$, then rectify.

Example

- ▶ Tableau:

$$T = \begin{array}{|c|c|c|c|c|} \hline 1 & 2 & 3 & 7 & 9 \\ \hline 4 & 5 & 10 & & \\ \hline 6 & 8 & & & \\ \hline \end{array}$$

- ▶ $I_1 = [1, 5]$, $I_2 = [6, 10]$:

$$T_{I_1} = T^{I_1} = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline \end{array}, \quad T^{I_2} = \text{rect} \left(\begin{array}{c} \begin{array}{|c|c|} \hline 2 & 4 \\ \hline \end{array} \\ \begin{array}{|c|} \hline 5 \\ \hline \end{array} \\ \begin{array}{|c|c|} \hline 1 & 3 \\ \hline \end{array} \end{array} \right) = \begin{array}{|c|c|c|} \hline 1 & 2 & 4 \\ \hline 3 & 5 & \\ \hline \end{array}.$$

Section 7

Main result: the two-row case $\lambda = (n, n)$

The two-row case

- $\lambda = (n, n)$:

$$c_{\lambda, \mu} < |\text{SYT}_{\lambda}(\mu)|,$$

for some $\mu \vdash 2n$.

- Refined Thrall subset: impose extra conditions on $\text{SYT}_{\lambda}(\mu)$.

Definition

$$\text{SYT}^=(2n) = \{ T \in \text{SYT}(2n) : T_{[n]} = T^{[n+1, 2n]} \}.$$

Example

$$T = \begin{array}{|c|c|c|c|} \hline 1 & 2 & 5 & 6 \\ \hline 3 & 4 & 8 & \\ \hline 7 & & & \\ \hline \end{array} \in \text{SYT}^=(8)$$

$$T_{[4]} = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} = \text{rect} \left(\begin{array}{c} \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 4 & \end{array} \\ \begin{array}{|c|} \hline 3 \\ \hline \end{array} \end{array} \right) = T^{[5, 8]}.$$

The bijection ξ and spin

Lemma

► *Bijection:*

$$\xi : \text{SYT}^=(2n) \longrightarrow \bigsqcup_{\substack{\lambda \vdash n \\ \mu \vdash 2n}} \text{SYT}(\lambda) \times \text{YDT}(\lambda^\square, \mu),$$

► $\xi(T) = (U, D): U = T_{[n]}, \text{wt}(D) = \text{sh}(T).$

► $\lambda^\square = (2\lambda_1, 2\lambda_1, 2\lambda_2, 2\lambda_2, \dots).$

► YDT: **Yamanouchi domino tableaux**.

Definition

► $T \in \text{SYT}^=(2n), \xi(T) = (U, D):$

$$\text{spin}(T) := \text{spin}(D) = V(D)/2,$$

► $V(D)$: number of vertical dominoes in D .

Main theorem

- ▶ $<$: any total order on all SYT.
- ▶ Define:

$$\text{SYT}_{(n,n)}^{<}(\mu) = \{ T \in \text{SYT}_{(n,n)}(\mu) : T_{[n]} < T^{[n+1,2n]} \},$$

$$\text{SYT}_{(n,n)}^{\text{spin}}(\mu) = \{ T \in \text{SYT}_{(n,n)}(\mu) : T_{[n]} = T^{[n+1,2n]}, \text{spin}(T) \equiv n \pmod{2} \}.$$

Theorem (Main result)

- ▶ *Schur expansion:*

$$\text{ch}(L_{(n,n)}) = \sum_{\mu \vdash 2n} |\text{SYT}_{(n,n)}^{<}(\mu) \sqcup \text{SYT}_{(n,n)}^{\text{spin}}(\mu)| s_{\mu}.$$

- ▶ *Refined $((n,n), \mu)$ -Thrall subset:*

$$\text{SYT}_{(n,n)}^{<}(\mu) \sqcup \text{SYT}_{(n,n)}^{\text{spin}}(\mu)$$

Example

- ▶ $n = 3, \lambda = (3, 3)$.
- ▶ $\text{SYT}_{(3,3)}^<(\mu) = \emptyset$ for all $\mu \vdash 6$.
- ▶ Eight tableaux $T \in \bigcup_{\mu \vdash 6} \text{SYT}_{(3,3)}(\mu)$ with $T_{[3]} = T^{[4,6]}$:

spin = 3

1	3	4	6
2	5		

T_1

spin = 1

1	3	6
2	4	
5		

T_2

spin = 1

1	3	6
2		
4		
5		

T_3

spin = 1

1	3
2	4
5	6

T_4

spin = 2

1	3	4	6
2			
5			

T_5

spin = 2

1	3	4
2	6	
5		

T_6

spin = 2

1	3	4
2	5	6

T_7

spin = 0

1	3
2	6
4	
5	

T_8

- ▶ $\text{spin} \equiv 3 \pmod{2}$: keep T_1, T_2, T_3, T_4 .

$$\text{ch}(L_{(3,3)}) = s_{(4,2)} + s_{(3,2,1)} + s_{(3,1,1,1)} + s_{(2,2,2)}.$$

Section 8

The bijection ξ

The bijection ξ in three steps

- ▶ $T \in \text{SYT}^=(2n)$.
- ▶ $U = T_{[n]}, S = \text{st}(T_{[n+1, 2n]}).$
- ▶ Three steps:

$$\text{SYT}^U(\mu/\lambda) \xrightarrow{\Psi_U} \text{LR}(\mu/\lambda, \lambda) \xrightarrow{\Omega} \text{LR}(\lambda * \lambda, \mu) \xrightarrow{\Phi_0} \text{YDT}(\lambda^\square, \mu),$$

- ▶ Pair with U :

$$\xi(T) = (U, (\Phi_0 \circ \Omega \circ \Psi_U)(S)).$$

- ▶ Ψ_U : van Leeuwen (2001)
- ▶ Ω : van Leeuwen (2001)
- ▶ Φ_0 : van Leeuwen (2000)

The bijection ξ

<https://jungw-s.github.io/thrall-2row-bijection-visualizer/>



Open problems

Open problem 1

- Describe $\text{spin}(T)$ **intrinsically** for $T \in \text{SYT}^=(2n)$, without ξ .

Open problem 2

- Solve Thrall's problem for rectangles $\lambda = (n^k)$, $k \geq 3$.

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Thank you!